Prompt Engineering

Prompt

- A prompt is a detailed set of instructions or guidelines given to a Large Language Model (LLM) to perform a specific task.
- It tells the model what to do, how to do it, and what kind of output is expected.

Prompt Engineering

- Prompt engineering is the process of designing and refining prompts to get the best possible results from an LLM.
- It involves creating task-specific prompts and improving them through trial and error (iterative process).

Prompt Elements

1. **Instruction**:
   The specific task or action you want the LLM to perform.
   Example: "Write test cases for login functionality."
2. **Context**:
   Background information or details that help the LLM understand the task better.
   Example: "The login functionality has two fields: username and password. The password must be at least 8 characters long."
3. **Input Data**:
   The data or information provided to the LLM to work with.
   Example: "Username: 'testuser', Password: 'password123'."
4. **Output Indicator**:
   The format or structure you want the output in.
   Example: "Provide the test cases in a table format with columns for Test Case ID, Description, Input, Expected Output, and Actual Output."

Example Prompt:

"*Write test cases for a login functionality. The login has two fields: username and password. The password must be at least 8 characters long. Use the following input: Username: 'testuser', Password: 'password123'. Provide the test cases in a table format with columns for Test Case ID, Description, Input, Expected Output, and Actual Output.*"

Prompt Engineering Techniques

1. **Zero-Shot Prompting**:
   Asking the LLM to perform a task without providing any examples. The model relies on its pre-trained knowledge.

   Example:
   "Write test cases for a login functionality with username and password fields."

2. **One-Shot Prompting**:
   Providing a single example to guide the LLM on how to perform the task.

   Example:
   "Write test cases for login functionality. For example, if the username is 'user1' and the password is 'pass123', a test case could be:
   - Test Case ID: 1
   - Description: Valid login
   - Input: Username: 'user1', Password: 'pass123'
   - Expected Output: Login successful
     Now, write test cases for a login functionality with username and password fields."

3. **Few-Shot Prompting**:
   Providing multiple examples to help the LLM understand the task better.

Example:
"Write test cases for a login functionality. Here are some examples:

- Test Case ID: 1
  Description: Valid login
  Input: Username: 'user1', Password: 'pass123'
  Expected Output: Login successful
- Test Case ID: 2
  Description: Invalid password
  Input: Username: 'user1', Password: 'pass'
  Expected Output: Login failed, password too short
  Now, write test cases for a login functionality with username and password fields."

Summary:

- **Prompt**: Instructions given to an LLM to perform a task.
- **Prompt Engineering**: Iteratively refining prompts to get better results.
- **Prompt Elements**: Instruction, Context, Input Data, Output Indicator.
- **Techniques**: Zero-shot (no examples), One-shot (1 example), Few-shot (multiple examples).